Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 30)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 1 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 1 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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0 2 . 1 Which of the following is the main reason that lead-based paint is a greater risk to children than to adults? Tick one box.

[1 mark]

A Children inhale more lead vapour ☐

B Children pick up dust / flakes and suck their fingers ☐

C Lead is more soluble in children's blood ☐

D Children spend more time near roads ☐

0 2 . 2 Explain how tri-butyl tin (TBT) in anti-fouling paint caused pollution even though only small amounts dissolved into the sea.

[3 marks]

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0 2 . 3 Iron is not toxic, yet iron-rich drainage from mine spoil heaps kills fish. Explain why, and describe how it is controlled.

[3 marks]

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0 2 . 4 Give **two** uses of lead that continue because they do not cause significant lead release into the environment.

[2 marks]

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0 2 . 5 Explain why heavy metals are described as neurotoxins.

[1 mark]

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0 3 . 1 Explain why abstraction from an aquifer containing 'fossil' (palaeo) water, such as the High Plains / Ogallala Aquifer in the USA, cannot be sustainable.

[3 marks]

0 3 . 2 Describe how the GRACE satellites are used to monitor aquifer depletion and give **one** advantage over monitoring boreholes.

[3 marks]

0 3 . 3 Describe **three** measures being introduced on the North China Plain to reduce aquifer depletion.

[3 marks]

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0 3 . 4 Suggest why large-scale afforestation on the North China Plain may reduce aquifer recharge even though trees reduce soil erosion.

[1 mark]

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0 4 . 1 Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

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0 4 . 2 Give **three** factors that affect the choice of disposal method for a solid waste.

[3 marks]

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0 4 . 3 Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill.

[2 marks]

Advantage .....

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Disadvantage .....

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0 4 . 4 Explain why mixed municipal waste is difficult to recycle.

[1 mark]

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0 5 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 5 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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0 5 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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0 6 . 1 Explain how each of the following environmental factors affects the rate of degradation of pollutants. Give a named example of a pollutant for each factor.

[6 marks]

Temperature .....

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Light .....

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Oxygen .....

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0 6 . 2 Explain how the presence of adsorbent materials such as clay can reduce the severity of pollution by heavy metals or radioactive caesium-137.

[2 marks]

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0 6 . 3 Explain why a fall in pH increases the harm caused by many heavy metals.

[2 marks]

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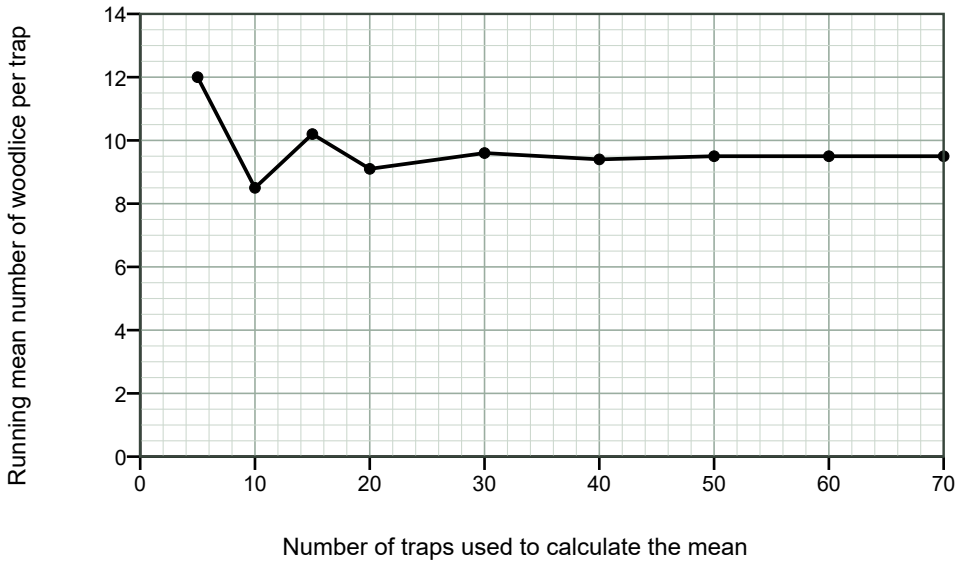
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0 7 Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



0 7 . 1 Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

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0 7 . 2 Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

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0 7 . 3 The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

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0 7 . 4 Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

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0 8 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

0 8 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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0 8 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

0 8 . 4 Give **three** other positive feedback mechanisms.

[3 marks]

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0 8 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

15

0 9 Table 1 shows the average annual radiation dose received by a person in the UK from different sources.

Table 1

Source	Percentage of average dose / %
Radon gas from the ground	50
Medical (X-rays, CT scans)	16
Gamma rays from rocks / buildings	13
Internal (food and drink)	9.5
Cosmic radiation	10
Occupational, fallout, nuclear discharges, consumer products	1.5

0 9 . 1 Explain why radon concentrations are high inside some houses in Devon and Cornwall.

[3 marks]

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0 9 . 2 Use Table 1 to calculate the percentage of the average dose that comes from artificial (human-made) sources.

[1 mark]

Answer ..... %

0 9 . 3 Suggest **two** reasons why the dose received by an individual may differ greatly from the average.

[2 marks]

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0 9 . 4 Explain why a risk:benefit analysis is needed for the use of X-rays in medicine, and give **two** reasons why the risks of ionising radiation are difficult to assess.

[4 marks]

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1 0 . 1 Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

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1 0 . 2 State what is meant by the **Leq 57 dB** contour used around UK airports.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2 Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS